

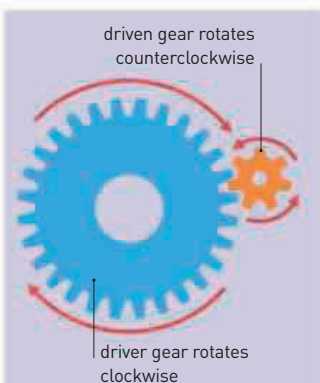


A medieval depiction of a Vitruvian undershot waterwheel. Operated with a hand lever, the buckets fill with water as the wheel rotates and the buckets dip into a water source. The water is deposited at the top.

“THE LAWS OF MECHANICS ARE FOUNDED ON THOSE OF NATURE, AND ARE ILLUSTRATED BY STUDYING THE MASTER-MOVEMENTS OF THE UNIVERSE ITSELF.”

Marcus Vitruvius Pollio, Roman architect and engineer, from *Ten Books on Architecture*, c.15 BCE

**THE ANTIKYTHERA MECHANISM IS A COMPLEX DEVICE** that shows the earliest understanding of gears. Dating from around 80 BCE, it was recovered in 1900 from a shipwreck off the Greek island of Antikythera. Made up of a series of bronze toothed dials and at least 30 gears, it is thought to have been used to **predict solar and lunar eclipses** and to track other



#### GEARS

Mentioned c.330 BCE by Aristotle, the Romans brought gears into common use during this period in waterwheels and hoists. Gears are made up of sets of interlocking toothed wheels. They work when a larger wheel engages with a small wheel and alters the speed of a driving mechanism.

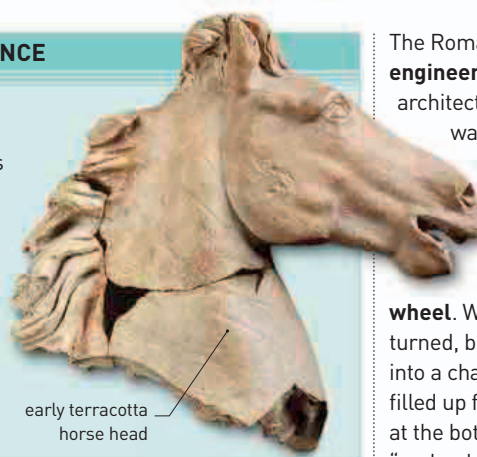
time cycles, such as the 19-year Metonic cycle—the basis for the ancient Greek calendar.

By the 1st century BCE the **Maya calendar** had developed a 5,125-year era known as the **Long Count**. Twenty *tun* (years) made a *katun*, 20 *katun* were a *baktun*, and 13 of these completed the whole era. The earliest known date inscribed in the Long Count system is December 9, 36 BCE; this is found on a stele at Chiapa de Corzo in Mexico. The Maya also used a 52-year Calendar Round, with two elements working in combination—the 260-day Tzolk'in calendar and the 365-day Haab.

Around 90 BCE **Posidonius of Apamea** (c.135–50 BCE) used the relative position of the star Canopus, seen from Alexandria and Rhodes, to **calculate the size of Earth**. His calculation was 240,000 stades, only slightly smaller than the estimate of **Eratosthenes of Cyrene** (see 250–100 BCE). Posidonius also calculated the size of the Moon and made a study of tides,

#### ROMAN VETERINARY SCIENCE

Roman interest in veterinary science sprang from the needs of farmers and also of the army, which had large cavalry units. In the army, specialists called *mulomedici* cared for military donkeys and horses. Around 45 CE the Roman writer Columella wrote extensively on the care and diseases of farm animals.



early terracotta horse head

relating them to the phases of the Moon.

Around this time, the Greek physician Asclepiades of Bithynia (c.129–40 BCE) put forward his idea of the **brain being the seat of sensation**. He developed a theory of disease based on the flow of atoms in the body, a doctrine he derived from the atomic theory of the 5th-century-BCE philosopher Democritus. His treatment methods were very subtle, prescribing baths and exercises. Perhaps less humane

were the practices of his follower Themison of Laodicea, who was the first recorded physician to use **leeches to bleed patients**.

The Roman writer Celsus (c.25 BCE–50 CE) produced one of the most important **texts on medicine**, *De Medicina*, an encyclopedic summary of medical knowledge of the time. In it, he gave accounts of the use of opiates for calming patients and laxatives to purge them. He also detailed many **surgical techniques**, including the removal of kidney stones and how to operate on cataracts (clouding of the lens in the eye).

The Romans also advanced **engineering** in this period. The architect Vitruvius (c.84–15 BCE) was the first to explain the use of siphons to lessen hydraulic pressure in pumps. He also described the **Vitruvian turning wheel**. When the wheel was turned, buckets emptied water into a channel at the top and filled up from a water source at the bottom. This type of “undershot” waterwheel had probably been invented earlier, but Vitruvius may have refined it to make it more effective.

Glassblowing was developed around 50 BCE in Roman-controlled Syria. **Glassmakers** obtained a more even flow by blowing **molten glass** through a tube (either freely or into a mold), rather than just pouring it. The **higher-quality glassware** that resulted led to the establishment of glassworks throughout the Roman Empire.



**Roman glass**  
The strong colors of this 1st-century CE vase from Lebanon are typical of the early Imperial period.

**3 MILLION THE DIAMETER OF THE SUN IN STADES, AS PER POSIDONIUS**

